Lecture 38 — When $Ax = b$ has no exact solution

Mon Nov 30

Last updated: October 29, 2026.

Boyd video (optional): Lecture 33 — least squares · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 12.1. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Look ahead: Wednesday turns the projection picture into normal equations, software fits, and positive definiteness; Friday's quiz / lab / SVD launch are on the schedule.

Learning objectives

By the end of class, you should be able to:

1. Recognize an inconsistent overdetermined system.
2. Formulate the least-squares objective $\min_x \|Ax - b\|$.
3. Interpret $\hat{b} = A\hat{x}$ as the **orthogonal projection** of b onto $C(A)$.
4. Interpret the residual $r = b - \hat{b}$ as orthogonal to $C(A)$ at a minimizer.
5. Distinguish an exact solution from a best approximate solution.

Warm-up

Consider the overdetermined system $Ax = b$ with

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}.$$

(a) How many equations? How many unknowns? Is A tall, wide, or square?

(b) Suppose someone guesses $x = (1, 0.5)$. Compute Ax and $Ax - b$. Is x an exact solution?

Fill in

Overdetermined / inconsistent. When there are more independent equations than unknowns, $Ax = b$ often has _____ exact solution. The right-hand side b may lie outside the column space $C(A)$.

Least-squares idea. Instead of insisting on $Ax = b$, choose $\hat{x}$ to make $A\hat{x}$ as close as possible to b:

$$\hat{x} \;=\; \arg\min_x \|Ax - b\|.$$

The vector $\hat{b} = A\hat{x}$ is the _____ attainable right-hand side in $C(A)$.

Projection picture. Among all vectors in the subspace $C(A)$, the closest one to b is the **orthogonal projection** of b onto $C(A)$. Least squares is exactly that geometry in disguise:

$$\hat{b} \;=\; \text{proj}_{C(A)}(b), \qquad \hat{b} = A\hat{x} \text{ for some } \hat{x}.$$

(So $\hat{x}$ are the coordinates of that projection in the columns of A.)

Recall — orthogonal vectors (Lecture 4). Two vectors u, v are **orthogonal** when their inner product is zero: $u^\mathsf{T} v = 0$ (a right angle). Same vocabulary as early in the semester; now one vector is the residual and the other lives in $C(A)$.

Residual. The **residual** is

$$r = b - A\hat{x} = b - \hat{b}$$

(some texts write $A\hat{x} - b$; same length $\|r\|$). At a minimizer, r is _____ to every vector in $C(A)$: for every column a_j of A,

$$a_j^\mathsf{T} r = _____.$$

(That stack of n equations is $A^\mathsf{T} r = 0$ — Wednesday's bridge to the normal equations.)

Exact vs approximate.

Situation	Meaning
Exact solution	$Ax = b$ for some $x \Leftrightarrow b \in$ _____ $\Leftrightarrow \hat{b} = b$
Least-squares solution	$\hat{b} = \text{proj}_{C(A)}(b)$; residual may be _____

If an exact solution exists, it is also a least-squares solution (residual zero — b already lives in $C(A)$).

Worked in class

W1. Inconsistent check. For the warm-up A and b, argue that no $x = (x_1, x_2)$ can satisfy all three equations simultaneously. (Hint: equations 1–2 force $x_2 = 1$; plug into equation 3.)

W2. Objective value. With the same A, b and trial $x = (1, 0.5)$, compute $\|Ax - b\|$ (Euclidean). Is this the minimum? (You need not prove optimality yet — just evaluate the objective.)

W3. Projection sketch. Sketch (in $\mathbb{R}^3$ as a cartoon, or think in words): $C(A)$ is a plane (or line) through the origin; b is off the plane; $\widehat{b} = \text{proj}_{C(A)}(b)$ is the foot of the perpendicular. Label $r = b - \widehat{b}$ and mark the right angle between r and the plane — same "$u^{\mathsf{T}}v = 0$" idea as Lecture 4, now between r and every direction in $C(A)$.

W4. Exact case. Change b to $b' = (1, 2, 3)$. Show that $x = (1, 1)$ solves $Ax = b'$ exactly. What is the residual? What is $\text{proj}_{C(A)}(b')$ in this case?

W5. Language check. In one sentence each: (i) "no exact solution," (ii) "$\widehat{b}$ is the projection of b onto $C(A)$."

Your turn

Y1. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \qquad x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Compute Ax and $r = b - Ax$. Is x an exact solution? What is $\|r\|$?

Y2. True or false, with a one-line reason: if $Ax = b$ is inconsistent, then no vector $\hat{x}$ can make $\|A\hat{x} - b\|$ finite.

Y3. In words: why do we call $\hat{b} = A\hat{x}$ the orthogonal projection of b onto $C(A)$?

Y4. A 4×2 matrix A has linearly independent columns, and $b \notin C(A)$. Can $Ax = b$ have an exact solution? Can it have a least-squares solution? What is $\text{proj}_{C(A)}(b)$ equal to in that case — b, or something else?

Answers (Your turn)

Y1. $Ax = (2, 1, 1)$; $r = (1, 0, 0)$; not exact. $\|r\| = 1$.

Y2. **False.** The residual norm is always finite; least squares still chooses an $\hat{x}$ that minimizes it (even when the minimum is positive).

Y3. It is the unique closest point in $C(A)$ to b, and the error $b - \hat{b}$ is orthogonal to every vector in $C(A)$ in the Lecture 4 sense ($u^{\mathsf{T}}v = 0$) — that is the definition of orthogonal projection onto $C(A)$.

Y4. Exact: **no** ($b \notin C(A)$). Least squares: **yes**. Here $\text{proj}_{C(A)}(b) = \hat{b} = A\hat{x} \neq b$ (something else — the foot of the perpendicular in $C(A)$).